13 · DNA, chromosomes and mutations

The words of this track, as it says them. Bold brown is the examiner.

DNA, chromosomes and mutations.

Describe the structure and function of DNA and RNA. Then define some of the ways genetic abnormalities occur, including mutation, genetic disease and aneuploidy.

Two checklist points in one track. Her checklist calls the DNA part a review from B N 1, and her Module 3 quizzes only touch it lightly, so here's a version that answers both.

The hook: DNA is the recipe book, a chromosome is one volume, a gene is one recipe, and a mutation is a typo.

DNA is deoxyribonucleic acid. It's a double helix: two strands of nucleotides twisted together, each nucleotide made of a sugar, a phosphate and a base. The bases pair up across the middle: adenine with thymine, and guanine with cytosine. Its function: it stores the genetic code, and it's the molecule of heredity, passing traits from one generation to the next.

RNA, ribonucleic acid, is single-stranded, with the sugar ribose, and uracil in place of thymine. Its job is to carry the code out of the nucleus, as messenger RNA, to the ribosomes, where proteins are made. That RNA part is the textbook review; her Module 3 bank doesn't ask it.

Now her words. A chromosome is a coiled piece of DNA that contains many genes. A gene is a segment of DNA that codes for a trait or a protein. Humans have forty-six chromosomes. A gamete, a sperm or an egg, has twenty-three, and the zygote has forty-six, half from the mother and half from the father. And in meiosis, crossing over, where paired chromosomes swap pieces at a chiasma, can separate genes that sit close together, the linked genes. That's one source of variation.

How genetic abnormalities happen. A mutation is a permanent change in the genetic structure of a gene: a change in the DNA code. A genetic disease is one caused by a faulty gene. Her examples: Huntington's, which is dominant, and cystic fibrosis and Tay-Sachs, which are recessive. Her quiz also notes that schizophrenia and Down syndrome are not simple recessive traits. And aneuploidy is an abnormal number of chromosomes, as in Down syndrome or Turner syndrome. That one has its own track.

One more, from her true or false: why does a fatal recessive disease like Tay-Sachs persist? Because carriers, the heterozygotes, don't have the disease, but they pass the allele on.

Once more. DNA, chromosomes and mutations.

DNA, deoxyribonucleic acid, is a double helix of nucleotides, with adenine paired to thymine and guanine to cytosine. It stores the genetic code and passes traits on. RNA is single-stranded, with ribose and uracil, and carries the code out to make proteins. A chromosome is coiled DNA with many genes, and a gene is a segment that codes for a trait

or a protein. A gamete has twenty-three chromosomes and the zygote forty-six, half from each parent. A mutation is a permanent change in a gene's DNA code. A genetic disease comes from a faulty gene, like Huntington's, cystic fibrosis or Tay-Sachs. Aneuploidy is an abnormal number of chromosomes.